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SUPERSEDING
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MILITARY SPECIFICATION

PRALIDOXIME CHLORIDE INJECTION

This specification is approved for use by Defense Personnel Support Center, Department of Defense, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Pralidoxime Chloride Injection in a pressure-activated automatic injector suitable for intramuscular administration (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-N-196	- Needles, Hypodermic, for Luer Syringes.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular Polyethylene Foam, Flexible.
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personne Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th Street, Philadelphia, PA 19101, by using the f-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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STANDARD

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting officer.)

2.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein:

U. S. DEPARTMENT OF HEALTH AND HUMAN SERVICES,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein.

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Physical characteristics.

3.1.1 Injector. Shall be a pressure-activated, automatic injector suitable for intramuscular administration of the contained medication. The injector shall have a length no greater than 6.0 inches and a diameter no greater than 0.780 inch. These exterior dimensions shall be controlled by appropriate testing of components prior to assembly. The injector shall have the needle end permanently identified by a black color. The safety mechanism shall be permanently identified by a grey color. Each end shall be capable of being identified by sight and by touch. The medication shall be contained in an ampule unit consisting of a glass container (Type I tempered borosilicate glass, USP), an injector needle, a rubber stopper to prevent loss of medication and function as a piston to drive out the medication, and a rubber stopper or diaphragm between medication and needle. The ampule unit shall be considered acceptable to the Food and Drug Administration for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

3.1.2 Needle. The needle used in the injector shall be 20, 21, or 22 gauge inclusive as defined in GG-N-196 when tested as specified therein.

3.2 Medication. The medication in the injector shall contain the ingredients listed in Table I.

TABLE I. Medication.

INGREDIENT	AMOUNT
Pralidoxime Chloride, USP $C_7H_9ClN_2O$	600 mg (plus a 10 percent overage)
Benzyl Alcohol, NF	40.0 mg
Aminoacetic Acid, USP	22.52 mg
Hydrochloric Acid, Reagent Grade or NF	qs pH 2.5
Water for Injection, USP	qs to 2.0 ml

3.2.1 Clarity. The injection shall be essentially clear and free from undissolved or any foreign particulate matter.

3.2.1.1 Color. The injection shall be colorless to a slightly yellow solution.

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3.2.2 Assay. Shall assay to contain not less than 270 nor more than 360 mg Pralidoxime Chloride per ml when tested as specified in 4.2.5. Shall assay to contain not less than 18 nor more than 22 mg Benzyl Alcohol per ml when tested as specified in 4.2.5. The concentration of Aminoacetic Acid shall be determined from the batch production records.

3.2.2.1 Reference standard. Referenced standard for Pralidoxime Chloride may be USP reference standard for validated house standard which has met the requirements specified in 4.2.5.2.

3.2.3 pH. The pH of the finished injection shall be between 2.0 and 3.0 when determined as specified in 4.2.6.

3.3 Ingredients.

3.3.1 Pralidoxime chloride. The pralidoxime chloride used in the manufacture of the injection shall comply with the tests, standards and requirements of the USP, including any supplements or revisions therein.

3.3.2 Benzyl alcohol. The benzyl alcohol used in the manufacture of the injection shall comply with the tests, standards and requirements of the NF, including any supplements or revisions thereto.

3.3.3 Aminoacetic acid. The aminoacetic acid used in the manufacture of the injection shall comply with the tests, standards and requirements of the USP, including any supplements or revisions thereto.

3.3.4 Hydrochloric acid. The hydrochloric acid used in the manufacture of the injection shall comply with the tests, standards and requirements of the ACS for Reagent Grade or NF, including any supplements or revisions thereto.

3.3.5 Water for injection. The water for injection used in the manufacture of the inspection shall comply with the tests, standards and requirements of the USP, including any supplements or revisions thereto.

3.4 Activation force. The injector shall activate when a force of 2.0 to 8.0 lbs is axially applied as specified in 4.2.4.1. The front end of the injector shall not become dislodged during activation.

3.5 Volume of medication and dispensing time. The injector shall dispense 2.0 + 0.2 ml of medication during a test period of no more than 4 seconds from the time of activation when tested as specified in 4.2.4.2.

3.6 Needle length after medication is dispensed. After the injector is activated and the medication is dispensed, the free (exposed) length of the neeedle shall be 18 to 23 mm when tested as specified in 4.2.4.3.

3.7 Rough handling and safety mechanism. The injector shall contain a safety mechanism capable of preventing accidental activation or dispensing of medication due to dropping, snagging, bumping, or other rough handling of packaged injectors removed from intermediate carton when tested as specified in 4.2.4.4. The injector shall not activate with the safety mechanism in place when a force of 20-22 lbs is axially applied as specified in 4.2.4.4.

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3.8 Ability to withstand high altitude (low pressure). When the injector is conditioned to 20 hours at an absolute pressure of no more than 196 mm of mercury and then for 4 hours at normal pressure, and this conditioning is repeated three times (total conditioning time of 4 days), the injector shall show no evidence of medication leakage when tested as specified in 4.2.4.5.

3.9 Sterility. The injection and all components and surfaces in contact with the medication shall be sterile. When the injector is activated and the medication is ejected, the liquid shall be sterile when tested in accordance with the USP. This test shall be performed after the injector has been conditioned as specified in 3.8. The medication and all components and surfaces in contact with the medication shall be sterile when tested as specified in 4.2.4.6.

3.9.1 Certificate of sterility. A certificate of sterility, listing all lot or control numbers, shall state that samples from each sterility lot have been tested in accordance with USP method and found to be sterile.

3.10 Pyrogens. The injection shall meet the USP requirements when tested as specified in 4.2.4.7.

3.11 Permanency of marking. Each injector shall be durably and legibly marked or labeled as specified. When tested as specified in 4.2.4.8, the marking shall remain entirely legible; any labels used shall be unaffected and shall remain entirely adhered to the injector.

3.11.1 Label information. Each injector label shall bear only the following information and shall be sized and placed to allow for inspection of the contents. The injector label of immediate container shall be light brown, and shall contain the following:

a. Product name: PRALIDOXIME CHLORIDE INJECTION, (Black) (The product name shall appear as the first item on the label.)

b. 300 mg/ml, 2 ml.

c. For use in NERVE AGENT POISONINIG ONLY (Black)

d. Instructions (Black type on white background that has been outlined in black)

(1) REMOVE GRAY SAFETY CAP (with direction arrow to safety area)

(2) PLACE BLACK END ON OUTER THIGH AND PUSH HARD TILL INJECTOR FUNCTIONS (with direction arrow to needle)

(3) HOLD FIRMLY IN PLACE FOR TEN SECONDS

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e. Lot number (Black)

f. Date of manufacture (Black)

g. Name and address of manufacturer and/or distributor (Black)

h. National Stock Number (Black)

i. The label shall bear any information required by FDA. In addition, the label may bear the trade name, and patent number. (Black)

j. A NATO light brown band centered on a white background shall be printed on the entire edge of the label next to the needle end.

3.13 Delivery. Not more than 3 months shall have elapsed from the date of manufacture to the date of delivery to Government.

3.13.1 Definition of date of manufacture. The date of manufacture is the date of filling or of sterilizing the final containers, or the date of manufacturer's or contractor's final quality approval of the final containers, which shall be not later than one month after the date of filling.

3.14 Workmanship. The injector shall be free from visible damage such as cracks, dents, and distortions and from foreign matter such as grease, oil, and dirt. The medication and the injector shall be free from any defects which detract from their appearance or may impair their serviceability. The manufacturer shall establish adequate controls, procedures and documentation to assure compliance with the workmanship requirement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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4.1.2 No company supplying any raw materials or injector components to the contractor or injector manufacturer will be considered an acceptable facility for the performance of any inspection requirements specified herein. Inspection and test requirements not specified herein but referenced by applicable documents in 2.1, 2.2 and 2.3 may be performed at the option of the contractor or injector manufacturer.

4.2 Quality conformance inspection.

4.2.1 Lot.

4.2.1.1 Lot for inspection. For purposes of this specification, a lot, batch or control is that single, uniform, and homogeneous quantity of the medication produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.2.1.2 Pharmaceutical lot or finished injectors. A lot shall consist of the injectors produced from one lot of the medication at one plant, under essentially the same manufacturing conditions. Sterility sublots may be defined, sampled, and tested in accordance with USP definitions. Retain samples of each sterility sublot will be held; 20 minimum from each sterility sublot.

4.2.1.3 Lot for water for injection, USP. A lot shall not exceed that quantity which is distilled from the still(s) into one container during a 24 period. As an alternate the water shall commence to be used within eight (8) hours after removal from the heated recirculating water for injection system.

4.2.2 Sampling. Where possible the sample shall be used for 2 or more tests.

4.2.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 as specified in 4.2.3.1.

4.2.2.2 For test. The samples for each test shall be selected in accordance with MIL-STD-105, Level S-4 using an AQL of 1.5 (percent defective), except as follows:

a. Samples for sterility test shall be taken as specified in USP with an acceptance number of zero.

b. Samples for pyrogens shall be taken as specified in 4.2.4.7 with an acceptance number of zero.

c. Samples for assay shall be taken as specified in 4.2.5.1 with an acceptance number of zero.

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4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample injectors and the preparation for delivery thereof shall be examined in accordance with MIL-STD-105 and the following classification of defects. Examination shall include, but shall not be limited to, the classified possible defects listed below:

TABLE II. Classification of defects.
(Injector)

CATEGORIES	DEFECTS
Critical:	AQL: No critical defects are allowed. Application of this requirement dictates an AQL of 0.065 or tighter.
1.	Markings or labeling of the injector incorrect, missing or illegible; Level II.
2.	Needle missing; Level I.
3.	Medication missing; Level I.
Major:	AQL 2.5 (percent defective)
101	Needle gauge not as specified; Level S-4.
102	Workmanship; Level II.
	AQL 1.0 (percent defective)
103	Solution not essentially clear and free from undissolved or foreign particulate matter. This check examination may be performed prior to the colorless glass cartirdge being inserted into the injector; Level II.

TABLE III. Classification of defects.
(Packaging (Section 5))

CATEGORIES	DEFECTS
Critical:	None defined.
Major:	AQL 4.0 (percent defective); Level II
101	Contents per shipping container not as specified.
102	Shipping container damaged.
103	Marking of intermediate and exterior containers incorrect, missing, or illegible.

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4.2.3.2 For tests. Samples selected in accordance with 4.2.2.2 shall be tested as specified in 4.2.4.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Activation force. Remove the safety mechanism from the injector. Place the injector in a suitable device capable of applying and measuring required loads. Apply a load axially to the injector with the least possible impact and increase the load until the injector is activated. Record the actual load at which the injector was activated. Examine the injector to determine if the front end has become dislodged during activation.

4.2.4.2 Volume of medication and dispensing time. Using suitable apparatus, activate the injector and determine the volume of medication dispensed by weight, corrected for specific gravity. Measure the volume of medication and to the nearest second the time between activation and the stop of the flow of medication from the needle.

4.2.4.3 Needle length after medication is dispensed. Using suitable apparatus, activate the injector. After the medication has stopped flowing from the needle, measure the length of the needle protruding from the injector to the nearest millimeter. As an alternate, the gauging of the needle on limit gauge for length shall be permitted.

4.2.4.4 Rough handling and safety mechanism. Drop injectors which have been removed from the intermediate cartons onto a concrete floor from a height of 4 feet. Repeat the drop three times. Examine the injectors for evidence that they would be unusable. Without removing the safety mechanism, attempt to activate the injectors by applying an axial force of not less than 20 lbs. to not more than 22 lbs.

4.2.4.5 Ability to withstand high altitude (low pressure). Place the injector in a suitable vacuum chamber and reduce the pressure to between 165 and 196 mm mercury absolute and maintain the pressure for 20 hours. Return the pressure to normal for 4 hours. Repeat the pressure cycle 3 times (total of 4 cycles). Remove the injector from the chamber and visually examine for evidence of medication leakage.

4.2.4.6 Sterility. Condition the injector as specified in 4.2.4.5. Remove the filled cartridge and rubber needle guard component from the injector. Penetrate the rubber needle guard with the needle. Eject the medication from the cartridge and test the ejected medication for sterility in accordance with the applicable portions of the sterility tests contained in the USP.

4.2.4.7 Pyrogens. Meets the USP requirements when 2.5 ml of injection is diluted to 33 ml and the diluted solution is injected at a dose of 1 ml, per kg.

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4.2.4.8 Permanency of marking. Rub the marking or labeling on the injector ten times with fingers using moderate pressure. Examine for legibility and permanency of marking or labeling.

4.2.5 Assays on final product. As an alternate to the assay methods specified below, the finished product may be assayed in accordance with the manufacturer's or contractor's NDA method.

4.2.5.1 Pralidoxime chloride.

SCOPE

This method is applicable to aqueous solutions containing about 330 mg of Pralidoxime Chloride per ml. The method is specific for Pralidoxime Chloride in the presence of degradation products such as 2-carboxy-N-methylpyridinium chloride, N methyl-2-pyridone, 2-carbamyl-N-methylpyridinium chloride, and 2-cyano-N-methylpyridinium chloride.

OUTLINE OF METHOD

The sample is diluted, first with purified water, and then with the aqueous acetonitrile mobile phase. The diluted solution is chromatographed on a reverse phase column (octadecylsilane bonded to micro-particles of silica). The amount of pralidoxime chloride in the sample is calculated from the area of the Pralidoxime peak in the sample preparation chromatogram, compared to the areas of Pralidoxime peaks in standard preparation chromatograms obtained concomitantly.

EQUIPMENT

Column - Length 250 mm, about 3 mm I.D., packed with Spherisorb^{R} ODS (5 micrometers).

Pump - Spectra-Physics Model 740-P, or equivalent.

Injector - Rheodyne Model 7210 with 50 ul loop, or Waters Associates WISP 710 auto injector, or equivalent.

Detector - Perkin-Elmer LC-75, or equivalent, or use at 300 nm.

Electronic integrator - Hewlett-Packard 3354B, or equivalent.

Membrane filters - Pore size 0.45 micrometer, Millipore^{R} HA, available from Millipore Corporation, Bedford, MA 01732, or equivalent.

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REAGENTS

Glacial acetic acid - Reagent grade.

Tetraethylammonium chloride (TEAC) - Eastman Organic Chemicals Catalog No. 3022, or equivalent, available from chemical supply houses.

Acetonitrile - "Distilled in Glass" grade, Burdick and Jackson Laboratories, Inc., 1953 S. Harvey Street, Muskegon, MI 49442, or equivalent.

Pralidoxime chloride reference standard - Of known strength (P).

Aqueous component - Add 0.7 gram of Tetraethylammonium Chloride and 1.0 ml of glacial acetic acid to a 1 liter volumetric flask and dilute to volume with purified water (this solution will be 0.004M in TEAC and 0.175M in acetic acid). Filter the solution through a 0.45 micrometer pore size membrane filter.

NOTE: The amount of Tetraethylammonium Chloride may be varied ± 50 percent to optimize chromatography.

Mobile phase - Mix 700 ml of acetonitrile with 300 ml of aqueous component. Degas in an ultrasonic bath for 10 minutes.

PREPARATION OF STANDARDS

1. Accurately weigh about 50 mg, 65 mg, and 75 mg (W_p) of Pralidoxime Chloride reference standard and transfer the weighed standards to separate 50 ml volumetric flasks.

   NOTE: If extensive decomposition is expected, a fourth portion of about 25 mg of reference standard may be weighed and transferred to a separate 50 ml volumetric flask.

2. Dissolve the weighed standards and dilute to volume with purified water.

3. Pipet 1.00 ml of each of the solutions from step (2) into a separate 50 ml volumetric flask and dilute to volume with mobile phase.

EQUIPMENT PREPARATION AND SYSTEM SUITABILITY TEST

1. Set the detector wavelength to 300 nm and inject 50 ul of the middle strength standard preparation. Adjust the flow rate of the mobile phase and the sensitivity of the detector so that the Pralidoxime peak elutes in about 5 minutes (flow rate 0.5 to 0.8 ml/minute), and is at least 50 percent full scale (about 0.32 absorbance units full scale sensitivity).

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2. After the chromatographic conditions have stabilized, inject 50 ul of the middle strength standard preparation three times, obtaining the chromatogram of each injection.

3. Obtain the area of the Pralidoxime peak in each chromatogram.

4. The system is considered suitable for use if the area of the Pralidoxime peaks are within 2.0 percent of each other.

PREPARATION OF SAMPLES

1. Pipet a volume (V) of sample, equivalent to about 330 mg of Pralidoxime Chloride, into a 250 ml volumetric flask and dilute to volume with purified water.

2. Pipet 1.00 ml of the solution from step (1) into a 50 ml volumetric flask and dilute to volume with mobile phase.

PROCEDURE

1. Inject 50 ul of each of the reference standard preparations and obtain the areas of the Pralidoxime peaks.

2. Inject 50 ul of the sample preparation and obtain the area of the Pralidoxime peak.

NOTE: If more than 10 sample preparations are to be chromatographed, the following injection sequence is recommended: Standard 1, 5 samples, Standard 2, 5 samples, Standard 3, etc. The number of sample injections between standards may be reduced but not increased.

CALCULATIONS

1. Obtain the least square linear regression line for the peak area vs. the Pralidoxime Chloride reference standard weight (W_p) for the standard preparations.

2. Obtain from the regression line the weight (W) corresponding to the peak area of Pralidoxime in the sample preparations.

3. Calculate the Pralidoxime Chloride content of the sample from the equation.

$$\text{mg Pralidoxime Chloride/ml} = \frac{(W)\ (5)\ (P)}{(V)}$$

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Where: W = weight of Pralidoxime Chloride corresponding to the area of the Pralidoxime peak in the sample preparation chromatogram (mg)

5 = dilution factor (250 ml/50 ml)

P = strength of Pralidoxime Chloride reference standard (decimal)

V = volume of sample taken for assay (ml)

4.2.5.2 Reference standard/validation of house standard.

SCOPE

This method is used to assay the Pralidoxime Chloride content of Pralidoxime Chloride raw material to be used as a house reference standard.

OUTLINE METHOD

An alkaline solution of Pralidoxime Chloride exhibits an absorbance maximum near 336 nm which is suitable for assay when compared with the absorbance of a reference standard solution.

SPECIAL EQUIPMENT

Spectrophotometer - Capable of isolating a bandwidth of 2 nm or less at 336 nm and of measuring absorbances in the 0.5-1.0 range with a precision of 0.01.

Cells - 1 cm pathlength, silica.

REAGENT

Sodium hydroxide - USP test solution.

STANDARDS

Pralidoxime chloride reference standard - USP reference standard.

Pralidoxime chloride reference standard solution -

1. Accurately weigh about 100 mg (Wr) of USP reference standard and transfer to a 100 ml volumetric flask. Dissolve and dilute to volume with purified water. Mix well.

2. Pipet 10.0 ml of this solution into a 100 ml volumetric flask and dilute to volume with purified water. Mix well.

3. Proceed as directed in PROCEDURE.

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SAMPLE PREPARATION

1. Accurately weigh about 500 mg (Ws) of sample and transfer to a 250 ml volumetric flask. Dissolve and dilute to volume with purified water. Mix well.

2. Pipet 5.0 ml of this solution into a 100 ml volumetric flask and dilute to volume with purified water. Mix well.

3. Proceed as directed in PROCEDURE.

PROCEDURE

NOTE: Once the sodium hydroxide T. S. has been added to the respective volumetic flasks, the samples must be read within 10 minutes.

1. Pipet 4.0 ml of the standard preparation and of the sample preparation into respective 50 ml volumetric flasks.

2. Dilute each of the above samples to about 40 ml with purified water.

3. Pipet 5.0 ml of sodium hydroxide T. S. into each flask.

4. Dilute the samples to volume with purified water and mix well.

5. Prepare a blank by diluting 5.0 ml of sodium hydroxide T. S. to 50.0 ml with purified water. Mix well.

6. Measure the absorbance of the standard solution (Ar) and the sample solution (As) at the maximum near 336 nm using the blank preparation to blank the instrument.

CALCULATIONS

$$\text{Percent Pralidoxime Chloride} = \frac{(As\ (Wr)\ (100)\ (62500)}{(Ar)\ (Ws)\ (12500)}$$

As = absorbance of sample solution.

Ar = absorbance of standard solution.

Wr = weight of standard (mg).

Ws = weight of sample (mg).

100 = factor to convert decimal to percent.

62500 = sample dilution factor.

12500 = standard dilution factor.

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4.2.5.3 Benzyl alcohol.

SCOPE

This method is applicable to aqueous solutions containing about 20 mg of benzyl alcohol per ml and variable amounts of Pralidoxime Chloride and its decomposition products.

OUTLINE OF METHOD

The sample is diluted with purified water. An aliquot of the diluted solution is shaken with benzene containing phenol as an internal standard. The benzyl alcohol is extracted into the benzene layer. A portion of the benzene layer is injected into a gas chromatograph. The peak area ratio for the benzyl alcohol and phenol peaks is obtained. A standard curve is obtained by chromatographing portions of three standards prepared concomitantly. The concentration of benzyl alcohol in the sample is calculated from the peak area ratio of the sample chromatogram compared to the standard curve obtained from the standard chromatograms.

EQUIPMENT

Column - Length 180 cm, about 4 mm I.D., packed with 5 percent Carbowax® 20M on Gas Chrom Q® support, 100 to 120 mesh.

Gas chromatograph - Equipped with a nitrogen gas supply, flame ionization detector, and peak area integrator.

Microsyringe - Capacity 10 ul.

Centrifuge tubes - Capacity 35 ml, screw-cap.

Mechanical shaker.

REAGENTS

Benzene - Reagent grade.

Phenol - Reagent grade.

Benzyl alcohol reference standard - Of known strength (P).

Internal standard solution - Prepare a solution of phenol in benzene at a concentration of about 2 mg per ml.

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EQUIPMENT CONDITION

Typical conditions for a Tracor MT-200 gas chromatograph are as follows:

Inlet temperature	- 195°C
Detector temperature	- 195°C
Column temperature	- 175°C
Nitrogen carrier gas flow rate	- about 35 ml/minute

These conditions may be varied if necessary to obtain the following approximate retention times:

Benzyl alcohol	- 4 minutes
Internal standard (phenol)	- 6 minutes

PREPARATION OF STANDARDS

1. Into three separate 50 ml volumetric flasks, weigh accurately about 30 mg (W_{r1}), 40 mg (W_{r2}), and 50 mg (W_{r3}) of benzyl alcohol reference standard.
2. Dissolve the weighed portions of reference standard and dilute to volume with purified water. Mix well.
3. Pipet 5 ml of each standard solution into separate 35 ml centrifuge tubes.
4. Pipet 5 ml of internal standard solution into each tube.
5. Cap the tubes tightly and shake for 30 minutes on a mechanical shaker.
6. Continue as directed under PROCEDURE.

PREPARATION OF SAMPLE

1. Pipet 1 ml of sample into a 25 ml volumetric flask.
2. Dilute to volume with purified water and mix well.
3. Pipet 5 ml of the solution from step (2) into a 35 ml centrifuge tube.
4. Pipet 5 ml of internal standard solution into the tube.
5. Cap the tube tightly and shake for 30 minutes on a mechanical shaker.

fied in 5.2.7.1 of PPP-C-186.

.3 Level C. Units shall be packaged in standard commercial containers e size and kind commonly used, which will afford the degree of protection red for shipment and use of the product for its intended purpose.

Packing. Packing shall be level A, B or C, as specified (see 6.1).

.1 Level A. Twenty-four intermediate packages (288 units) shall be

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PROCEDURE

1. Make consecutive injections of 1 ul portions of the top (benzene) layer of one of the standard preparations until the peak area ratios agree within 2 percent for two consecutive injections.

2. Inject a 1 ul portion of the top (benzene) layer of each standard and sample preparation.

3. Obtain the peak area ratio (benzyl alcohol peak area/phenol peak area) for each standard and sample preparation chromatogram.

CALCULATIONS

1. Using the weights (W_{r1}, W_{r2}, W_{r3}) of the reference standard portions as the abscissa, and the respective peak area ratios as the ordinate, determine the slope (m) and the y-intercept (b) of a standard calibration curve for benzyl alcohol.

2. Calculate the benzyl alcohol content of the sample from the equation:

$$\text{mg benzyl alcohol/ml} \quad \frac{(R-b)\ (p)}{(m)\ \ (2)}$$

Where:

R = Sample preparation peak area ratio
b = y-intercept of calibration curve
m = slope calibration curve (mg^{-1})
P = strength of the reference standard (decimal)
2 = dilution factor (ml), from $\frac{(50\ ml)\ (1\ ml)}{(25\ ml)}$

4.2.6 pH determination. The pH should be between 2.0 and 3.0 when determined potentiometrically at 25°C ± 2°C using the USP method.

5. PACKAGING

5.1 Packaging. Packaging shall be level A, as specified (see 6.1).

5.1.1 Unit. One (1) injector, as specified, constitutes one unit.

5.1.2 Level A. Two (2) injectors shall be placed with points down in pouch formed from nominal 1/8 inch thick polyethylene foam conforming to a type II, class 1 of PPP-C-1752. The pouch shall have inside dimensions of 6-1/4 + 1/8 inch by 2-3/8 ± 1/8 inch and the opening shall be in one end along the smaller dimensions.

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5.1.2.1 Intermediate packaging. Fifty pouches of two injectors each (one hundred injectors) packaged as specified in 5.1.1, shall be intermediately packaged in a fiberboard box conforming to Grade W5c of PPP-B-636. The box shall have approximate inside dimensions of 12-5/8 inches by 11-1/2 inches by 7-1/4 inches. Layers of pouches shall be separated by fiberboard pads formed from material conforming to minimum Grade W5c of PPP-F-320. Motion of contents within the box shall be prevented by the use of cushioning material conforming to type II, class A of PPP-C-843. The box shall be closed as specified in the appendix of PPP-B-636.

5.2 Packing. Packing shall be level A or C, as specified (see 6.1).

5.2.1 Level A. Two intermediate packages of injectors prepared as specified in 5.1.2.1 shall be packed in a fiberboard box conforming to grade V3c of PPP-B-636 for type 2 load of 50 to 100 pounds. The box shall have approximate inside dimensions of 23-7/8 inches by 13-1/4 inches by 8-3/4 inches. Motion of contents shall be prevented by the insertion, where needed, of fiberboard pads formed from material conforming to grade W5c of PPP-F-320. The box shall be closed and reinforced by filament-reinforced tape banding as specified in the appendix of PPP-B-636.

5.2.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Marking.

5.3.1 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed. Lot (control) number, contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129 except that the date of manufacture shall be shown in lieu of the date packed. Lot (control) number shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specificaton.
b. National Stock Number.
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specificaton covers the following item:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6505-01-125-3248	PRALIDOXIME CHLORIDE INJECTION, Automatic, 300 mg per ml, 2 ml

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6.3 Right to retest. The contractor or manufacturer of the injector may retest the lot provided the Government is notified of any failure. Notice to the Food and Drug Administration shall constitute notice to the Government.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
DLA-DM

Project No. 6505-2139

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